Measuring What We Mean
Construct Validity When Measures Are Cheap

A position paper on construct-identified inference in the age of AI

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Status. The companion package `cvprofiles` 3.0.1 is publicly released (PyPI, 2026-08-14) and used here as an instrument. This version reports a human-only application of construct-identified inference to patience and trust at two resolutions—countries and country-demeaned demographic cells—under networks and thresholds frozen before slacks were inspected. The 2026-08-10 seven-measure patience freeze, which included open-weight log-probability scores, is retained as an appendix pilot and is not the headline evidence. Empty admissible sets, wide uncertainty bands, and disagreement across resolutions are reported as findings.

Statement of significance

Scientists rarely observe constructs such as patience or trust directly; they work through measures. Artificial intelligence has made candidate measures cheap, so the scarce task is deciding which scores warrant the intended interpretation and which scientific conclusions survive that choice. We compose construct validity, multiverse reporting, and partial identification around four objects: a construct C, a finite menu $\mathscr{M}$, a theory-derived validity argument $\mathscr{R}(C)$, and a downstream estimand $\beta(m)$. Survivors form $\mathscr{M}^*$; the image $\mathscr{B}^*$ reports remaining disagreement among those who may speak. An unrestricted menu of country-level patience measures sign-switches on a development estimand; the same objects, under a declared network, leave a fragile singleton. The same grammar applied to trust, and to within-country cells, does not reproduce that singleton. The paper's result is that disagreement—across measures, resolutions, and the screen versus the estimand—is the measurement conclusion, not a prelude to a later experiment.

Abstract

Artificial intelligence has sharply reduced the cost of producing candidate measures of latent constructs. A researcher can now measure patience or trust with a validated survey module, a proxy item, a composite, a prompted language model, or a trained policy. Abundance creates a selection problem: which measures warrant the interpretation they are given, and which scientific conclusions survive reasonable measurement choice?

We develop construct-identified inference around four objects: a construct C, a declared menu $\mathscr{M}$, a theory-derived validity argument $\mathscr{R}(C)$, and a downstream estimand $\beta(m)$. The validity argument screens the menu into an admissible set $\mathscr{M}^*$; the image $\mathscr{B}^* = \{\beta(m) : m \in \mathscr{M}^*\}$ reports which results survive measurement choice within the declared contract. If $\mathscr{R}$ is empty, that image is the unrestricted specification-curve range. Selection is made prospectively falsifiable by holding out restrictions, and sampling fragility induced by data-dependent screening is reported additively as an uncertainty band, not as a confidence set for the headline range.

The framework is implemented in the open, model-free package `cvprofiles` 3.0.1. On a frozen seven-measure patience menu the empty-network special case recovers the sign-switching range $[-0.219, 0.402]$. A Campbell–Fiske 2×2 written as ordinary restrictions retains no instrument: at country level the canonical WVS trust item tracks GPS patience more closely than GPS trust. The confirmatory application is human-only and two-resolution. At the country level, only GPS patience survives an education and risk-discriminant screen ($n = 41$), with plug-in coefficient 0.402 on log GDP per capita after an education control and an uncertainty band $[0.079, 0.563]$ that is much wider than the point. Several WVS trust facets survive

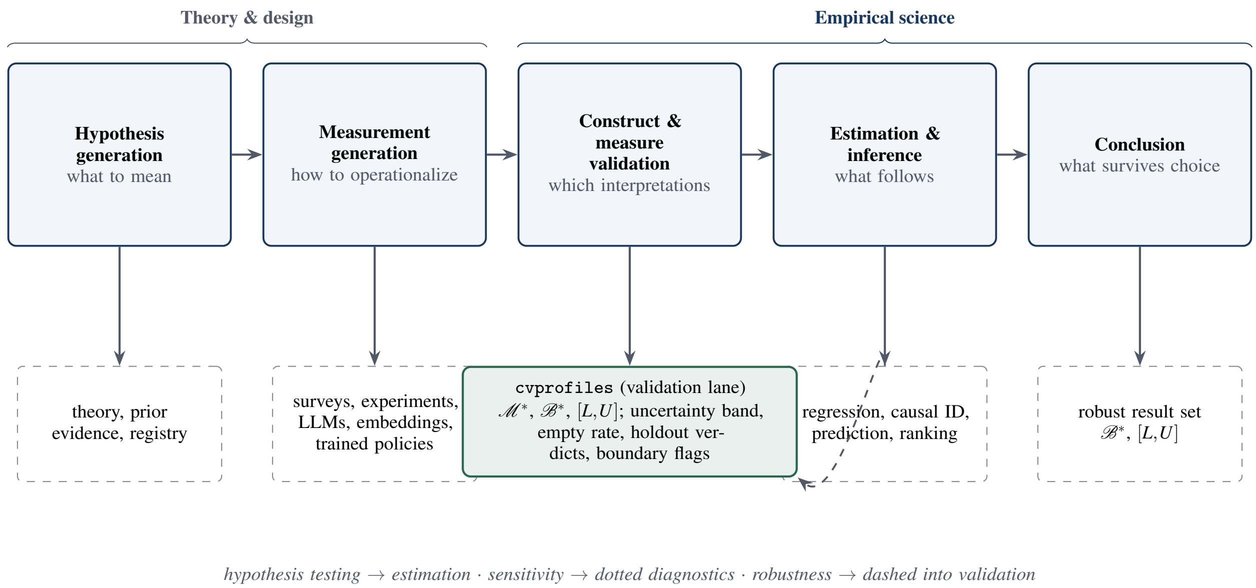


Figure 1: A systems view of empirical knowledge production. AI primarily expands the measurement-generation stage; this paper disciplines the bridge from construct validation to estimation and conclusions. Solid arrows denote artifacts, dashed denote researcher-owned commitments, and the diagnostics that monitor the validation lane are reported separately from the headline outputs.

institutional and inequality bars ($n = 35$) with range $[0.107, 0.391]$, but the coverage band crosses zero. Inside countries, after country demeaning, thrift (Q13) is a weak admissible representation of GPS patience and perseverance (Q14) is not; no WVS trust measure clears the predeclared education bar, even though several multi-item facets still co-move with GPS trust. The result is not a settled measure of either construct. It is that the same objects produce different speaker lists at different resolutions, and that those disagreements are what the composition is for.

Keywords: construct validity; measurement uncertainty; partial identification; operationalization; multiverse; AI measurement; construct-identified inference

1. The inferential problem when measures become cheap

1.1 Science is a pipeline

Empirical science is a pipeline. Hypothesis generation proposes constructs and candidate operationalizations; measurement generation maps inputs into scores; construct validation asks whether those scores support the interpretation they are given; estimation uses the scores in downstream analyses; and a conclusion stage reports what has been learned. Each stage has its own tools and its own failure modes, and the stages compose: a conclusion is only as strong as the weakest link in the chain (Figure 1). For most of the history of empirical social science, the binding constraint sat at the measurement stage: producing a new measure was costly, and validation effort concentrated on a few carefully constructed instruments. Artificial intelligence relaxes that constraint. Prompted language models, log-probability elicitation, embeddings, and trained policies can each produce a candidate score for the same construct at low marginal cost [26, 34], and the supply of candidate scores grows quickly. What does not grow is the evidence that any particular score column measures the construct. In particular, a standard error computed conditional on a chosen score column—however small—cannot establish that the column measures the construct: sampling variability and construct validity are different objects, and no amount of the former substitutes for the latter. When measures become cheap, validation becomes the scarce stage of the pipeline.

1.2 The measured failure mode

The change is not hypothetical; practice has been audited. Baumann et al. replicate annotation tasks from published social-science studies and show that plausible LLM prompt choices produce incorrect downstream conclusions for roughly 31% of hypotheses [2]. Desai, Card, and Jacobs audit LLM-based measurement in flagship social-science journals and find validation practices inconsistent and often limited relative to the centrality of the measurements [15]. Kim et al. document that LLM errors are correlated far beyond chance—same-wrong-answer agreement of 42% against a 13% chance baseline, with better models *more* correlated, not less [31]. Those audits document undeclared menus and correlated generator error; they are not, as reported, failed restrictions against a human criterion, and they are not evidence that $\mathscr{B}^*$ is empty. Two implications follow. First, agreement across models is weak evidence of validity: models that share training pipelines can share mistakes, and consensus may reflect correlated error rather than construct-relevant signal. Second, engineering quality is not a substitute for validation. Yin, Vu, and Persico show that LLM-based occupational-exposure scores—inputs to a large applied literature—are unstable across model choices [50]. The scale is measurable: in Hall's hand-coded classification of the PolMeth 2026 program, 26% of presentations were core generative-AI, and annotation plus measurement accounted for about 60% of those [22]. Measurement generation has arrived in the pipeline; validation practice has not kept pace.

1.3 The missing composition

The audits above share a feature: each documents a failure inside a single stage of the pipeline. What is missing is the composition—an account of how measurement, validation, and estimation combine into a scientific conclusion, and of what a robustness claim means once measures are cheap. Without such an account, two reflexes dominate. One declares a result fragile because it shifts under any new operationalization, treating every plausible score column as equally entitled to enter the analysis. The other declares a result robust because a single operationalization was retained. Both reflexes confuse the menu of candidate measures with the evidence about the menu. The claim this paper works from is boxed below.

CORE CLAIM. A scientific conclusion should not be declared fragile because it changes under operationalizations that fail a declared validity argument, nor robust because a single operationalization was retained. The relevant robustness set is the image of the downstream estimand over the surviving set: $\mathscr{B}^* = \{\beta(m) : m \in \mathscr{M}^*\}$.

The boxed claim relocates the burden of robustness: stability is required over measures that survive a declared validity argument, not over all plausible operationalizations. The rest of this section makes the objects precise and states the contributions.

1.4 Four objects

The framework needs four objects. A *construct* C is the scientific concept of interest, defined by its intended interpretation rather than by any single column. A *menu* $\mathscr{M}$ is the finite set of candidate measures the researcher declares. A *validity argument* $\mathscr{R}(C)$ encodes what a measure of C should do: convergent and discriminant implications derived from the construct's nomological role, written as moment inequalities $\mathbb{E}[g_r(m,V;\tau_r)] \geq 0$ over auxiliary variables V, with researcher-owned thresholds τ_r. A *downstream estimand* $\beta(m)$ is the scientific quantity computed when measure m is used. The validity argument screens the menu: a measure survives when its sample slack $s_r(m)$ clears every restriction,

$$\mathscr{M}^* \;=\; \big\{m \in \mathscr{M} : s_r(m) \geq 0 \text{ for every } r \in \mathscr{R}(C)\big\}, \tag{1}$$

and the surviving set $\mathscr{M}^*$—not a single retained operationalization—is the object of inference. Remaining measurement choice is then propagated into the result: the robust result set is the image of the estimand over survivors,

$$\mathscr{B}^* \;=\; \big\{\beta(m) : m \in \mathscr{M}^*\big\}, \tag{2}$$

summarized for a scalar estimand by its range,

$$[L,U] \;=\; \Big[\min_{m\in\mathscr{M}^*} \beta(m),\; \max_{m\in\mathscr{M}^*} \beta(m)\Big]. \tag{3}$$

Equations (1)–(3) define the admissible set, the robust result set, and the headline range. Validity here is whether a score warrants the intended interpretation of C, operationalized as nonnegative slack on a declared $\mathscr{R}(C)$; robustness is whether β survives remaining measurement choice inside $\mathscr{M}^*$; the uncertainty band of Section 3.5 is selection uncertainty, not a confidence interval. The construct-identified result set is deliberately conditional—on the construct, the menu, the validity argument, the thresholds, and the sampling frame. Change any component and $\mathscr{M}^*$, $\mathscr{B}^*$, and $[L,U]$ are different objects; every component must therefore be declared and auditable rather than inferred from the data.

1.5 Four contributions

The contributions follow in deliberately conservative order. *(i) Validity-constrained measurement robustness.* $\mathscr{B}^*$ is the image of the estimand over measures that survive a declared validity argument, not an unrestricted multiverse; fragility and robustness are therefore attributed to the right cause. *(ii) A falsifiable selection rule.* Restrictions or units can be held out prospectively, and validity-guided selection can fail—against held-out evidence and against a random-selection baseline. *(iii) Separation of headline from diagnostics.* The headline range is reported separately from additive sampling diagnostics: an uncertainty band (selection uncertainty, not a confidence set), an empty-replicate rate, and boundary attribution. *(iv) An auditable, model-free implementation and a two-resolution application.* The method ships in the released package `cvprofiles` 3.0.1 [12]. The confirmatory application places human measures of patience and of trust under one contract at two resolutions, with leftover restrictions, additive sampling diagnostics, and two nested special cases computed on the same objects.

1.6 Roadmap

The remainder of the paper locks two commitments. Novelty is claimed only in the operational composition of three existing traditions, in a selection rule that can fail, and in a released implementation. The application is evidence about two constructs at two resolutions, not a general standard journals should require and not a measure of culture. Section 2 situates the argument among three traditions; Section 3 formalizes the objects, including the empty-network and Campbell–Fiske special cases; Section 4 documents the released implementation; Section 5 reports the two-resolution application; Section 6 states what the framework does and does not claim; Section 7 concludes.

2. Three traditions, one missing layer

2.1 Construct validity supplies the restrictions

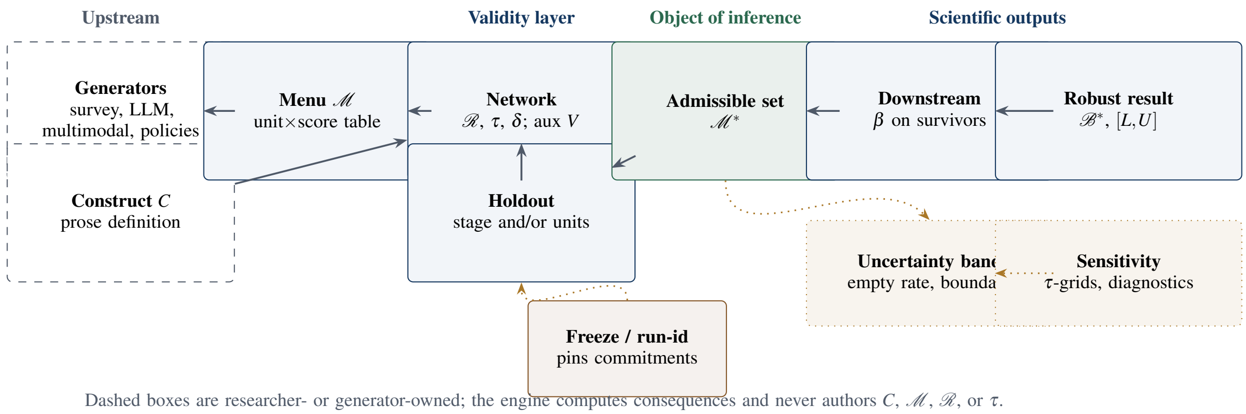
Construct validity is the oldest and most mature of the three traditions. Cronbach and Meehl introduced the nomological network: a construct earns its meaning through the laws and observable implications in which it participates, and validation consists of testing those implications [11]. Campbell and Fiske sharpened the empirical content into convergent and discriminant criteria, operationalized through the multitrait–multimethod matrix [7]. Messick and, later, Kane reframed validity as the defensibility of score interpretation and use—a property of interpretations and uses, not of instruments [36, 30]. Borsboom and colleagues added a realist caution: a measure is valid for a construct only if the construct exists and causally produces the score [5]. This paper takes the observable-implication strand [11, 7]. Kane's view is the pair $(C, \beta(m))$, not an extra screen on m; Borsboom's causal-production standard is not required for the plug-in admissible set, so an empty $\mathscr{M}^*$ is a failed nomological test, not a failed existence claim. Computational social science has begun importing this vocabulary. Li et al. formalize the proxy presumption—treating embeddings as measures without evidence—and propose a construct-validity protocol with discriminant, incremental, and predictive checks [42]. Halterman and Keith stage validation for LLM-based coding systems [23]. Licht et al. show that the elicitation protocol, such as pairwise comparison over direct scoring, materially changes what LLM-based measures record [33]. These contributions share a unit: each validates a *procedure*—a scoring method, an item, an instrument—against a criterion. Campbell and Fiske already place several measures in one matrix; the matrix is evidence about each cell. What that literature does not do is take the survivors as the domain of a downstream estimand and report $\mathscr{B}^* = \{\beta(m) : m \in \mathscr{M}^*\}$.

2.2 Multiverses expose operationalization uncertainty but do not adjudicate meaning

A second tradition quantifies how conclusions depend on analytic choices. Multiverse analysis enumerates a large space of defensible specifications and reports the distribution of results across it [46]; specification-curve analysis supplies a reporting discipline for the same idea [45]. At the measurement layer the logic is striking: Hanel and Zarzeczna enumerate 262,143 operationalizations of well-being and show that conclusions about the well-being of religious groups depend on which operationalization one examines [24]. The multiverse is an exploratory surface: it maps where conclusions live under alternative choices. It does not, by itself, adjudicate which choices carry the intended interpretation. Del Giudice and Gangestad make the point sharply: candidate measures differ in validity and reliability, and treating them as interchangeable branches of a multiverse can itself mislead [13]. The relevant robustness set is therefore not the image of the estimand over every plausible operationalization; it is the image over the measures that survive a declared validity argument. An unrestricted multiverse describes the space of analytic choice; the image of β over $\mathscr{M}^*$, not over $\mathscr{M}$, is the robustness object.

2.3 Partial identification supplies the set logic, not the validity argument

The third tradition supplies the formal set logic. Partial identification asks what can be learned about a parameter from data and maintained assumptions alone, reporting identified sets rather than point estimates [35, 38]; confidence sets for partially identified parameters extend the logic to sampling uncertainty [29, 47]. The connection to measurement is recent and explicit. In the 2026 NBER Summer Institute Methods Lectures, *Estimation and Inference with AI-Generated Data*, Dell and Rambachan frame the nomological network as partial identification under theoretically motivated moment restrictions—Dell on measurement and construct validity, Rambachan on multiple measurements and data fusion—and explicitly call for principled, widely accepted methods for construct validity [14]. The nearest formal neighbor is Chen, Rambachan, and Tamer, whose restrictions are benchmark-informed bounds on error correlation for a downstream parameter [10]. The present framework differs in the object of inference: our restrictions are menu-level nomological moment inequalities, the surviving set $\mathscr{M}^*$ is the object, and the selection rule is held-out falsifiable. Other recent work occupies adjacent territory without a surviving set: Ogut and Yin derive curvature bounds from multiple nonlinear measurements [39]; Messing shows that naive standard errors in LLM evaluation pipelines can be 40–60% too small—a diagnosis about evaluation scores, not construct measures [37]; Canen and Enamorado correct regressor measurement error by split-sample procedures—a correction, not a selection [8]. None performs menu-level validation with a surviving set and a falsifiable selection rule.



Dashed boxes are researcher- or generator-owned; the engine computes consequences and never authors C, $\mathscr{M}$, $\mathscr{R}$, or τ.

Figure 2: Core objects and epistemic ownership of construct-identified inference. The admissible set $\mathscr{M}^*$ is the object of inference; $\mathscr{B}^*$ and $[L,U]$ are downstream scientific outputs computed only over survivors; sampling and sensitivity diagnostics live in a separate dotted lane; and a freeze/run-id pin sits between researcher commitments and engine compute.

2.4 AI makes the old problem large

The fourth development is not a tradition but a shock. AI measurement generators—prompted elicitation, log-probability scoring, persona steering [16], preference fine-tuning [41, 40], and synthetic-agent pipelines such as SCA [9]—expand the supply of candidate scores at low marginal cost. Each technology enters $\mathscr{M}$ only as a map on the same declared units; pipelines that replace the sample, including some SCA-style designs, are not menu expansions under Definitions A.1–A.3. Engineering sophistication is not construct validity: a fluent generator can produce a score column that fails every nomological implication of the construct it purports to measure. The correct discipline is to keep generators and validators separate. Generators expand the menu; they do not adjudicate it. That separation, long standard for conventional measures, becomes the central discipline when measures are cheap. AI does not create the measurement problem; it makes an old problem large.

2.5 Synthesis

Each tradition contributes one piece. Psychometrics supplies the restrictions and the interpretation-based conception of validity. The multiverse tradition supplies the demand that conclusions be reported across choices rather than under a single one. Partial identification supplies the set logic in which a surviving set and its image are well-defined. The composition those traditions do not jointly supply is the map $\mathscr{R}(C) \to \mathscr{M}^* \to \mathscr{B}^*$. A holdout that makes screening prospectively falsifiable, and a freeze that makes it auditable, are protocol commitments; they are not implied by the map. That composition—the ordering of established ideas rather than a new taxonomy—is the contribution of this paper.

3. Construct-identified inference

This section fixes the objects, the screen, the falsification designs, and the sampling diagnostics.

3.1 Primitives and notation

Every object of the framework has a formal counterpart; Table 1 collects them and Figure 2 assembles them into the workflow. A *construct* C is the scientific concept of interest, defined in prose by its intended interpretation and independent of any score column. A *measurement function* $m : \mathscr{X} \to \mathbb{R}$ maps available inputs to a unit-level score, or to a vector followed by a *declared* scalarization. The *menu* $\mathscr{M} = \{m_1, \ldots, m_J\}$ is the finite set of candidates declared by the researcher. *Auxiliaries* V are observed variables used to evaluate validity implications; they are not the construct. The *nomological network* $\mathscr{R}(C)$ encodes those implications as moment restrictions with substantive thresholds τ_r, following the classical validation tradition [11, 7] and its recent partial-identification reading [14]. The admissible set $\mathscr{M}^*$ retains the measures that pass every restriction; $\beta(m)$ is the downstream target; $\mathscr{B}^*$ is its image over $\mathscr{M}^*$, with scalar summary $[L,U]$; and $G(D,C)$ denotes a measurement-generation technology producing candidate measures from evidence D and construct C. The owner column of Table 1 states the division of labor: commitments are authored upstream; the engine computes their consequences.

Table 1: Objects of construct-identified inference. The owner column records who authors each object: the researcher or theory, the measurement generator jointly with the researcher, or the engine itself. The engine computes consequences of commitments; it does not author the construct, the menu, the restrictions, or the thresholds.

Object	Definition	Scientific owner
C	Construct defined in prose by its intended interpretation, independent of any score column	researcher / theory
m	Measurement function $m : \mathscr{X} \to \mathbb{R}$, or a declared scalarization; maps inputs to a unit-level score	generator + researcher
$\mathscr{M} = \{m_1, \ldots, m_J\}$	Finite declared menu of candidate measures	researcher / theory
V	Auxiliaries: observed variables used to evaluate validity implications	researcher / theory
$\mathscr{R}(C)$; τ_r	Nomological restrictions with substantive thresholds	researcher / theory
$\mathscr{M}^*$	Admissible set: measures passing every restriction up to tolerance δ	computed
$\beta(m)$	Downstream target functional evaluated at measure m	researcher / theory
$\mathscr{B}^*$; $[L,U]$	Image of β over $\mathscr{M}^*$; min–max scalar summary	computed
$G(D,C)$	Measurement-generation technology: evidence D and construct C produce candidate measures	generator + researcher

3.2 Restrictions as observable implications

A validity restriction must be a decision rule with a prespecified direction. We write each restriction $r \in \mathscr{R}(C)$ as a population moment inequality,

$$r : \qquad \mathbb{E}\big[g_r\big(m(X), V; \tau_r\big)\big] \geq 0, \tag{4}$$

where τ_r is the *substantive threshold*: a researcher-owned commitment justified by theory, prior literature, or a preregistered tolerance, and *not* a parameter estimated by the admissibility procedure. Weak inequalities are the normal form of one-sided validity content (“at least this association,” “no more than this contamination”). A numeric gloss fixes the reading: $\tau_{\text{conv}} = 0.39$ means that any measure whose population correlation with the convergent criterion falls below 0.39 is denied the interpretation attached to C, regardless of how well it predicts Y. Let $s_r(m)$ denote the sample slack of restriction r on measure m (positive when satisfied). For a declared tolerance $\delta \geq 0$, the admissible set is

$$\mathscr{M}^*(C; \mathscr{M}, \mathscr{R}, \delta) = \big\{m \in \mathscr{M} :\ s_r(m) \geq -\delta \ \ \forall r \in \mathscr{R}(C)\big\}. \tag{5}$$

Passing is conditional evidence: $m \in \mathscr{M}^*$ means m is consistent with the declared interpretation—not that m is the construct, that the menu is exhaustive, or that m is valid for uses the tests do not represent. An empty admissible set is a valid finding: it rejects the pair $(\mathscr{M}, \mathscr{R})$ as an operationalization of C, not C itself. A poor restriction can reject a useful measure; a weak network can admit a poor one. The framework disciplines these commitments by making them explicit and auditable—it does not make them disappear.

The screen is deliberately not a classical joint hypothesis test: there is no likelihood, no null distribution, and no prior; the engine computes transparent moments against declared thresholds and reports which measures clear them. The scientific content is the intersection of the restrictions, not an aggregate statistic. Concrete convergent, discriminant, and predictive tests fit this form [42, 23], in contrast to restrictions that bound measurement-error correlation structure [10].

3.3 From admissible measures to scientific conclusions

Researchers usually care about an effect, association, ranking, or forecast rather than the score itself; let $\beta(m)$ denote the result when measure m is used. Once the admissible set is defined, the conclusions licensed under the validity argument are the image

$$\mathscr{B}^*(C; \mathscr{M}, \mathscr{R}) = \big\{\beta(m) : m \in \mathscr{M}^*\big\}, \tag{6}$$

with the transparent scalar summary

$$[L,U] = \Big[\min_{m \in \mathscr{M}^*} \beta(m),\ \max_{m \in \mathscr{M}^*} \beta(m)\Big], \tag{7}$$

defined whenever $\mathscr{M}^* \neq \emptyset$.

The contrast with an unrestricted multiverse makes the ordering concrete [45]. If $\mathscr{R} = \emptyset$, then $\mathscr{M}^* = \mathscr{M}$ and $[L,U]$ is the specification-curve range over the full menu: every candidate is entitled to speak, and remaining disagreement is reported rather than screened. That object is a named special case of (7), not a missing network. Suppose five candidate measures deliver β values 0.10, 0.12, 0.35, -0.04, and 0.38. Reported over the full menu, the conclusion ranges from -0.04 to 0.38: sign-switching, and fragile on its face. If the validity argument retains only the two measures with $\beta(m) = 0.35$ and

$\beta(m) = 0.38$, the construct-identified range $[0.35, 0.38]$ is narrow, and the conclusion survives measurement choice. If all five survive, every candidate meets $\mathscr{R}(C)$ but admissible measures disagree on β, so no single estimand conclusion survives measurement choice within the declared menu. The multiverse domain is $\mathscr{M}$; the construct-identified domain is $\mathscr{M}^*$. $\mathscr{B}^*$ is not an unconditional identified set for the latent construct; it is the set of results produced by measures that survive a declared validity argument [35, 38].

The same equations recover two classical objects. If $\mathscr{M}^*$ is treated as known, endpoint inference for $[L, U]$ is ordinary inference for extrema over a fixed finite set, and a Bonferroni union of intervals is a conservative cross-check in the spirit of Imbens and Manski [29]. If each g_r is a Chen–Rambachan–Tamer restriction on error correlation [10], the identified set defined by (4)–(7) is their set. Campbell and Fiske's monotrait–heteromethod and heterotrait–monomethod cells [7] are likewise instances of (4): a convergent floor τ_{conv} is a corr_min restriction against a partner column, and a discriminant ceiling τ_{disc} is a corr_zero restriction. The paper computes both special cases on frozen scores in Section 5.3.

3.4 Falsifying the validity rule

A validity framework can become circular: inspect available evidence, write restrictions after seeing which columns would pass, and report that they pass. The two designs below test different claims about a *frozen* contract; they do not constrain how $\mathscr{R}$ was written. A validity argument chosen after inspecting the same scores it will screen remains circular even if a later split is labeled a holdout (Table 2).

Restriction-stage split. Partition the prespecified restrictions into

$$\mathscr{R} = \mathscr{R}_S \cup \mathscr{R}_H, \qquad \mathscr{R}_S \cap \mathscr{R}_H = \emptyset, \tag{8}$$

and admit on $\mathscr{R}_S$ only:

$$\mathscr{M}_S^* = \{m \in \mathscr{M} : \ m \text{ passes every } r \in \mathscr{R}_S\}. \tag{9}$$

The held-out restrictions $\mathscr{R}_H$ then ask whether the survivors satisfy validity implications that never voted on admission. Holdout verdicts are scientific *findings* about the selection rule; under a pure stage split they do not redefine the headline set, which is defined by $\mathscr{R}_S$ unless the protocol says otherwise.

Units split. Partition the units into training and hold frames. Admission uses training units only; the hold frame demands compliance of the selected measures. The robust set is

$$\mathscr{M}^*_{\text{robust}} = \mathscr{M}^*_{\text{select}} \cap \{m : \ m \text{ passes the hold-frame screen}\}, \tag{10}$$

and, when the protocol so defines, the headline range is the image of β on $\mathscr{M}^*_{\text{robust}}$, with β evaluated on the full pooled sample. Empty robust sets are valid findings.

Comparative claim. Let $L_H(m)$ be a declared loss over the held-out restrictions. The falsifiable prediction is not that every survivor passes every unseen test; it is that construct-guided selection improves unseen validity relative to competing rules:

$$\mathbb{E}\big[L_H(m) \mid m \in \mathscr{M}_S^*\big] \ < \ \mathbb{E}\big[L_H(m) \mid m \in \mathscr{M}_B^*\big], \tag{11}$$

where $\mathscr{M}_B^*$ is selected by a baseline rule—single-anchor fit, best predictive fit, an LLM judge, or random or convenience selection as the floor. Failure is informative: if survivors do no better than random, the network has not demonstrated predictive scientific content. Pre-data discipline fixes the partition, δ, the baseline rule, and the seed before any holdout data are examined; the freeze/run-id contract hash-pins the network, anchors, stage split, and holdout unit list before holdout scoring [12].

3.5 Sampling uncertainty after data-dependent admission

The headline object remains the plug-in range (7): the image of β on the estimated $\mathscr{M}^*$, with no model and no shrinkage. Because $\mathscr{M}^*$ is estimated, the following reporting layer is *mandatory and additive*:

(i) **Uncertainty band.** A units-resampling bootstrap recomputes slacks, $\mathscr{M}^*$, and the endpoints in every replicate (seeded); report per-side quantiles over non-empty replicates. The resampling unit must be declared so that every slack and every $\beta(m)$ is a function of the resampled units; in the country profiles the unit is the country, and in the cell profiles the unit is the demographic cell. Label the object an *uncertainty band*: selection uncertainty on the pooled sample, not a confidence interval and not a holdout-robustness band. The fixed-selection literature takes the admissible set as given [29, 47]; here selection is estimated.

Table 2: Two falsification designs. They answer different questions and are never conflated; the protocol states which design (if any) redefines the headline set.

Design	Admission uses	Holdout asks	Headline consequence
Restriction-stage split ($\mathscr{R} = \mathscr{R}_S \cup \mathscr{R}_H$)	$\mathscr{R}_S$ only; $\mathscr{R}_H$ never votes on admission	Whether survivors satisfy unseen validity implications; verdicts are findings	Selection set $\mathscr{M}_S^*$, unless the protocol says otherwise
Units split (training / hold frames)	Training units only	Whether selected measures comply on held-out units	Robust set $\mathscr{M}^*_{\text{robust}}$; headline range on the robust set when the protocol so defines

(ii) **Empty-replicate rate.** The share of replicates with empty $\mathscr{M}^*$ is reported explicitly; the band is conditional on non-empty admissible sets. If that share is large, the empty-replicate rate is the primary sampling-fragility summary.

(iii) **Boundary attribution.** Flag measures whose absolute slack margin satisfies $|\text{margin}_m| \leq \kappa \cdot \text{SE}_m$ ($\kappa = 2$ default): near-threshold admissions and near-miss rejections are fragile; far-rejected measures are not.

(iv) **Conservative cross-check (optional).** Compare a units bootstrap that re-forms $\mathscr{M}^*$ in every replicate with the same bootstrap that freezes $\mathscr{M}^*$ at the original estimate; only that contrast isolates admission–endpoint coupling from ordinary endpoint sampling error. A Bonferroni union computed on a fixed $\mathscr{M}^*$ measures interval conservatism, not coupling.

Under uniform consistency of the slacks and the target functional, and a separation condition excluding measures exactly on the admissibility boundary, the plug-in admissible set and range converge in probability to their population counterparts (Appendix A). The documented failure regime is coupling: a near-boundary measure that is also β-extreme couples admission with the endpoints, because its inclusion or exclusion moves the range directly.

3.6 Scope and composition

The framework covers finite, scalarized menus: every candidate reduces to a declared scalar (or scalarized) object, so survey items, composites, and trained policies enter symmetrically. It is level-agnostic: a menu exists at every level of the latent hierarchy—item scores, composite scores, population aggregates—and the same screen applies at each level. Composition with causal analysis is direct: if a design with assumptions $\mathscr{A}$ yields an identified set $\Theta(m;\mathscr{A})$ for each fixed measure, the measurement-robust causal object is

$$\bigcup_{m \in \mathscr{M}^*} \Theta(m;\mathscr{A}), \tag{12}$$

the union over the admissible set. Construct identification determines which measures may carry the interpretation; causal identification proceeds within the survivors. Neither step substitutes for the other.

4. An auditable implementation

`cvprofiles` is the open Python implementation of the measurement-selection layer [12]. Its design goal is modest: accept a frozen unit-by-measure score table, a researcher-authored set of restrictions, thresholds, and a downstream functional; return a full measure-by-restriction profile, the admissible set, the robust result set, and the sensitivity diagnostics. The engine implements the state machine

$$\texttt{SCORE} \longrightarrow \texttt{RESTRICT} \longrightarrow \texttt{IDENTIFY} \longrightarrow \texttt{REPORT}, \tag{13}$$

and accepts four plain-text inputs—scores, roles, network, and target specification—producing slacks, $\mathscr{M}^*$, $\mathscr{B}^*$, a survivors-only range, holdout verdicts, and the additive diagnostics of Section 3.5. Frozen runs hash the score matrix, network, and target specification into a reproducibility record, and an empty $\mathscr{M}^*$ is an exit-success scientific finding rather than an exception. The engine is deliberately score-agnostic and model-free: it contains no language model, searches no prompt space, and writes no theory. This separation is a paper–package contract: the package is released on PyPI (version 3.0.1, 2026-08-14) and used here as an instrument. Version 3.0.1 adds a named `empty_R` contract so that $\mathscr{R} = \emptyset$ is expressible and fail-loud rather than a silent admit-all; default `empty_R=false` is omitted from the network hash, so pre-3.0.1 freezes remain bit-stable. Software cannot resolve disagreements about what patience or trust means; it can make those disagreements inspectable. Competing researchers can author different networks, thresholds, or menus and see exactly which commitment changes the conclusion—a feature, not a hidden optimizer.

5. Two constructs, two resolutions

The framework is applied to patience and trust, each at two resolutions, under networks and thresholds frozen before slacks were inspected. The menus are human-only: GPS country scores, WVS Wave 7 items and composites, and a seeded noise control. Open-weight log-probability scores appear only in the 2026-08-10 appendix pilot. Country β is the standardized OLS coefficient of the measure on log GDP per capita with a country-mean education control; cell β is the correlation of the measure with the matching GPS cell mean after country demeaning. Headline admission uses the restriction-stage split of Section 3.4: leftover restrictions never vote on $\mathscr{M}_S^*$. Every confirmatory run reports an uncertainty band, an empty-replicate rate, and a τ-grid as additive diagnostics ($n_{\text{boot}} = 1000$, $\alpha = 0.10$, seed 20260814).

5.1 Constructs, frames, and menus

Patience is time preference as operationalized by the Global Preference Survey, whose module was designed by selecting and weighting survey items for predictive validity against incentivized experimental measures and validated across countries [18, 19]. A valid measure should rise with human capital and should not collapse into risk preference. Trust is the belief that other people have good intentions, or that most people can be trusted. GPS trust is a single experimentally validated item; WVS Q57 is the canonical survey item; in-group, out-group, and institution facets are distinct objects of trust [20, 44]. A valid measure should co-move with institutional quality and move opposite income inequality [1, 4]. When GPS trust sits in the menu it is a peer candidate, not a bar.

The country frame is the inner join of GPS, WVS Wave 7, and a pinned WDI snapshot, after a respondent floor of 30 and after masking WVS missing codes -1 through -5 without imputation. Patience is observed for 41 countries; trust, which also requires WGI rule of law and a gini series, for 35. Cells are sex-by-age-band means (six bands from 18–24 through 65+) built separately in GPS individual files and in WVS, then inner-joined, with a minimum of 20 respondents per cell and at least six cells per country: 480 cells in 42 countries. The confirmatory cell estimand country-demeans every column that enters $\mathscr{R}$ or β before SCORE. A pooled (undemeaned) cell run remains on disk as a contrast; it is not the within-country conclusion.

The patience country menu is GPS patience, WVS Q13 (thrift), Q14 (perseverance), their declared composite $z(\text{Q13}) + z(\text{Q14})$, and noise. The trust country menu is GPS trust, Q57, in-group, out-group, institution, the mean of the three multi-item facets (Q57 excluded), and noise. Cell menus drop the GPS column from $\mathscr{M}$ and use it only as the β outcome, so recovery cannot be tautological.

5.2 Networks

Patience at the country level is screened by $\text{Corr}(m, \text{q275\_mean}) \geq 0.20$ (select) and $|\text{Corr}(m, \text{risktaking})| \leq 0.35$ (select), with leftover rank-monotonicity in education at 0.15. The education floor sits well below the published country association of patience with years of schooling [18]; the discriminant bar admits the published patience–risk range of 0.230 [18] to 0.358 [25] and still rejects $r \gtrsim 0.6$. Trust at the country level is screened by $\text{Corr}(m, \text{rule\_of\_law}) \geq 0.30$ and a negative corr_sign on gini at 0.10, with leftover education at 0.20. There is no corr_min against GPS trust, because that column is in the menu. Cell networks keep only the education implication, now on country-demeaned cell education, with leftover rank-monotonicity; GDP, gini, and rule of law do not vary within country and are not used.

5.3 Nested special cases

On the frozen seven-measure patience score table of the 2026-08-10 pilot, setting $\mathscr{R} = \emptyset$ in `cvprofiles` 3.0.1 admits the full menu and returns $[L, U] = [-0.219, 0.402]$, which is exactly the minimum and maximum of the seven β values. The composition claim of Section 3.3 is therefore a computed result, not a verbal nesting. The same table already showed that an unrestricted multiverse of patience operationalizations sign-switches on the development estimand.

Campbell and Fiske's 2×2 is written as eight ordinary restrictions on a peer menu of GPS patience, WVS child-qualities, GPS trust, and Q57, with $\tau_{\text{conv}} = 0.30$ and $\tau_{\text{disc}} = 0.35$ ($n = 41$). The convergent floor sits at a published survey–behavioral trust bound and below the Falk et al. GPS–WVS trust Spearman of 0.49 [18]; the discriminant ceiling admits their country Corr(patience, trust) of 0.190. Both the classical per-instrument retain set and the engine's global $\mathscr{M}^*$ are empty. The binding cells are not subtle: GPS patience versus child-qualities is $r = -0.252$; GPS trust versus Q57 is $r = 0.284$, just under the floor; Q57 versus GPS *patience* is $r = 0.825$. Partialling log GDP per capita and education leaves that last association at 0.706. The inequalities are Campbell–Fiske cells, so the psychometric bridge is formally recovered. Substantively the 2×2 rejects every instrument: at country level the canonical WVS trust item is closer to GPS time preference than to GPS "best intentions." That is a finding about the named special case, not a reason to lower τ after seeing slacks.

Table 3: Confirmatory two-resolution profiles (`cvprofiles` 3.0.1, seed 20260814, $n_{\text{boot}} = 1000$). Country β is standardized OLS on log GDP per capita with an education control; cell β is the correlation with the matching GPS cell mean after country demeaning. Coverage is the additive uncertainty band at $\alpha = 0.10$. Pooled (undemeaned) cell runs are omitted: they are not the within-country estimand.

Profile	n	$\mathscr{M}^*$	$[L,U]$	Coverage	Empty-rep.
Patience, country	41	{GPS patience}	0.402	$[0.079, 0.563]$	0.297
Trust, country	35	{Q57, in, out, composite}	$[0.107, 0.391]$	$[-0.016, 0.639]$	0.002
Patience, cells	480	{Q13}	0.245	$[0.174, 0.335]$	0.170
Trust, cells	480	$\emptyset$	—	—	1.000

5.4 Country profiles

Table 3 collects the four confirmatory profiles. Only GPS patience survives the country patience screen. Its education slack is $+0.275$ and its risk-discriminant slack is $+0.075$; leftover monotonicity on the *full* 41-country frame is $+0.233$. Q13, Q14, and the composite fail education hard (slacks -0.533, -0.777, -0.722) and have negative β on log GDP (-0.166, -0.195, -0.219). The plug-in range is the singleton 0.402. The uncertainty band is $[0.079, 0.563]$, and 29.7 percent of bootstraps empty $\mathscr{M}^*$. The τ-grid empties the set at $\lambda = 0.5$ because the discriminant bar tightens below the sample $|r|$ of patience with risk. The country singleton is therefore selection-fragile: the plug-in point is not a narrow scientific conclusion.

Trust at the country level is the opposite pattern. GPS trust fails rule of law (slack -0.176) and leftover education; institution trust fails the same rule-of-law bar (slack -0.425) and has negative β (-0.092). Q57, in-group, out-group, and the multi-item composite survive. The plug-in range $[0.107, 0.391]$ is the image of those four coefficients. The coverage band $[-0.016, 0.639]$ crosses zero, so the plug-in interval cannot be advertised as a narrow positive development gradient. Leftover education is passed by Q57, in-group, and out-group and failed by the composite (slack -0.052). The tool's holdout pass rate is 0.75 against a random-subset mean of 0.52 at $k = 1$; that is not a strong win over chance at this leftover test.

5.5 Within-country cells

Country-demeaning the 480 cells inverts both pooled-cell stories, which is why the pooled runs are not confirmatory. Q13 was rejected in the undemeaned table because richer countries have both more education and lower thrift endorsement. Inside countries the sign flips: education slack $+0.051$, leftover monotonicity $+0.060$, and β versus GPS patience $+0.245$, with coverage $[0.174, 0.335]$ and an empty-replicate rate of 0.17. Q14 stays anti-aligned (slack -0.564, $\beta = -0.326$). The composite is Q13 dragged under the bar by Q14. Q13's education slack is only $+0.051$; $\lambda = 1.5$ empties $\mathscr{M}^*$. At the resolution the design claimed, thrift is a weak admissible representation of GPS patience and perseverance is not. The country-level rejection of the WVS patience *menu* remains, because Q13 still fails corr_min(q275_mean) across countries. Resolution disagreement is the result.

Every trust facet fails the education bar after demeaning, including at $\lambda = 0.5$. The empty set is not a close call. Q57's pooled cell survival was between-country: after demeaning, β versus GPS trust is $+0.052$. In-group, out-group, institution, and the composite still recover GPS trust inside countries (0.224, 0.211, 0.285, 0.318) without tracking within-country education. The stated admission bar and the GPS-recovery estimand disagree. Under the predeclared education restriction, no WVS trust measure is admissible at the within-country cell resolution; several multi-item facets nonetheless co-move with GPS trust, and Q57 does not.

5.6 What the application is allowed to mean

Validity decides who may speak. The range reports leftover disagreement among those who may. The band reports that the speaker list itself is noisy. Leftover restrictions ask whether unseen implications still agree.

Read that way, the application does not deliver a preferred measure of patience or of trust. An unrestricted patience menu sign-switches; the validity screen leaves a GPS singleton whose sampling fragility the band refuses to hide. WVS thrift and perseverance are not interchangeable proxies: inside countries one weakly recovers GPS patience and the other points the wrong way. Trust's country survivors clear external institutional bars that GPS trust itself fails, and the same education implication that admits Q13 inside countries empties the entire trust cell menu. Q57 tracks GPS patience at country level and does not recover GPS trust inside countries. Those are disagreements the composition is designed to make inspectable. They are not a license to move τ, to promote the 2026-08-10 posture-(a) freeze, or to call the menus a measure of culture.

Four concerns travel with every reading. The country samples are small ($n = 41$ and $n = 35$), so units-split holdout is power-limited and was not used as headline. Thresholds remain researcher-owned commitments; their provenance is auditable and the τ-grid is reported, but they are still judgment. Cell country-demeaning was a dated estimand amendment after a diagnostic pooled-versus-demeaned correlation check, not a post-hoc rescue of θ. Anchors and outcomes are separated by construction—GDP is β-only; GPS trust is never a bar when it is in the menu—but circularity is a standing risk rather than a solved problem.

The framework remains modular: a new generator can join a menu without rewriting the network, and engineering sophistication is not construct validity [43, 41, 40]. That modularity is not a license to treat the present human-only application as an AI-measurement result.

6. What follows—and what does not

6.1 What an empty or wide set means

An empty $\mathscr{M}^*$ rejects the pair consisting of the current menu and the validity argument on the stated sampling frame; it does not refute the existence of the construct. A wide $\mathscr{B}^*$ says the scientific conclusion remains sensitive even after the validity screen; a narrow $\mathscr{B}^*$ says the conclusion is robust across the surviving operationalizations. All of these outcomes are scientifically interpretable, and none should be tuned away by relaxing δ after seeing the result.

6.2 The hierarchy of uncertainties

The paper does not claim to discover a new category called measurement uncertainty. Psychometrics, metrology, latent-variable models, and multiverse work have long separated measurement from sampling and analytic choices. The claim is compositional: construct validity determines which measurement alternatives deserve scientific standing; measurement robustness propagates the unresolved choice among them; specification robustness can then vary analytic decisions conditional on each admissible measure; and sampling inference quantifies repeated-sample uncertainty within that design. The layers answer different questions and should not be collapsed into one standard error.

6.3 Beyond social-science measures

Benchmark scores are measurement functions too: they map model behavior into quantities interpreted as reasoning, safety, or knowledge. A family of task families can therefore be treated as a menu, with held-out task families and cross-method evidence supplying a validity network (Appendix C). The same logic applies to medical risk scores, remote-sensing proxies, and administrative indices. Domain-specific validity evidence changes; the composition does not.

6.4 Researcher-owned validation gates in automated science

Automation can expand the menu, compute scores, and execute a frozen network. It should not silently redefine the construct, rewrite the network after seeing results, or treat correlated model outputs as independent evidence. A credible automated-science architecture therefore needs a researcher-owned, contestable validation gate—one that can itself fail, and whose commitments can be compared across research teams (Appendix D). The most direct failure mode is Goodharting: a generator trained explicitly against the known restrictions. The defenses are procedural rather than magical—freeze the network and thresholds before final scoring, preserve negative controls, reserve restrictions or units, and keep generator provenance separate from validity evidence.

6.5 What this application decides—and what remains

The empirical results of Section 5 are not a preliminary agenda waiting for future experiments to decide whether construct-identified inference has bite; they are direct evidence of that bite on two consequential constructs across two resolutions. First, on country-level patience, the unrestricted specification curve spans $[-0.219, 0.402]$, while the nomological screen isolates the validated instrument and exposes that its point estimate (0.402) is sampling-fragile ($[0.079, 0.563]$) under units resampling. Second, on within-country demographic cells, the screen discriminates among semantically related proxies: thrift (Q13) survives as a weak representation of time preference ($\beta = 0.245$), while perseverance (Q14) is anti-aligned. Third, on trust, the framework shows that single-item generalized trust (Q57) tracks development and patience at the country level but fails to recover GPS trust inside countries, whereas multi-item facets co-move with GPS trust without tracking within-country education.

What remains is not justifying the necessity of construct validity, but extending its empirical scale:

(i) **Multimodal and LLM generators.** Evaluate generative language-model representations against these same frozen human networks, requiring the adapter provenance cards of Appendix E.

(ii) **High-dimensional demographic panels.** Scale cell-mean and individual-level gradient evaluations to broader survey intersections where within-country variation provides rich identification.

(iii) **Pre-registered construct registries.** Build open libraries of nomological networks, threshold justifications, and candidate menus so that researchers can contest measurement definitions transparently.

6.6 Open implementation and community

The registry of restriction types, target functionals, and generators is deliberately small, and its gaps are known: stability is schema-only, slacks are never learned, and coupling theory is deferred to a companion methods paper. The contribution we invite is concrete—new restriction types, β-functionals, and measurement generators—not a leaderboard.

7. Conclusion

Science does not become easier merely because candidate measures become cheap. Abundance moves the bottleneck from producing an operationalization to justifying one. The appropriate response is not to rank human surveys above AI, or fine-tuning above prompting, but to make competing measures answer to the same construct-level evidence. The framework asks five questions in order: What do we mean? How might we measure it? What should a credible measure do? Which measures survive? Which scientific conclusions survive with them? Formally,

$$C \longrightarrow \mathscr{M} \xrightarrow{\mathscr{R}} \mathscr{M}^* \longrightarrow \mathscr{B}^*.$$

The novelty is not construct validity, multiverse analysis, partial identification, or sampling inference in isolation; it is their operational composition at the measurement layer. When measures are cheap, validity becomes the scarce resource.

A. First-principles formalization

A.1 Primitives

Definition A.1 (Measurement function and menu)**.** *A* measurement function *is a map $m : \mathscr{X} \to \mathbb{R}$ (or to a finite choice simplex) producing a score or response distribution from available inputs $X \in \mathscr{X}$; a map to a vector followed by a* declared *scalar reduction is part of the measure's definition. A* menu *is a finite collection $\mathscr{M} = \{m_1, \dots, m_J\}$ declared by the researcher.*

Definition A.2 (Nomological restriction)**.** *Given auxiliaries V and a threshold $\tau_r \in \mathbb{R}$, a* restriction *is a functional inequality $\mathbb{E}[g_r(m(X), V; \tau_r)] \geq 0$. The* sample slack *$s_r(m)$ is a sample analogue of the left-hand side; the sign convention is $s_r(m) \geq 0$ when the restriction is satisfied.*

Definition A.3 (Admissible set and admissible-result set)**.** *For tolerance $\delta \geq 0$,*

$$\mathscr{M}^* = \big\{ m \in \mathscr{M} : \; s_r(m) \geq -\delta \;\forall r \in \mathscr{R} \big\}.$$

For a target functional $\beta : \mathscr{M} \to \mathbb{R}$,

$$\mathscr{B}^* = \big\{ \beta(m) : m \in \mathscr{M}^* \big\}, \qquad [L, U] = \big[\min \mathscr{B}^*, \, \max \mathscr{B}^* \big] \quad (\mathscr{M}^* \neq \emptyset).$$

Definition A.4 (Menu-level validity)**.** *Inference is* menu-level *when the object of scientific interest is $\mathscr{M}^*$ (and functionals of $\mathscr{M}^*$), not a single preferred m. Per-measure protocols [42, 23] validate individual operationalizations; they are complementary inputs to a menu, not substitutes for menu-level inference.*

A.2 Maintained assumptions

Assumption A.1 (Menu coverage)**.** *$\mathscr{M}$ contains the operationalizations of scientific interest for the research question; coverage is argued from literature and design, never proven.*

Assumption A.2 (Restriction validity)**.** *Each g_r is a correct testable implication of the construct definition and auxiliaries V; restrictions are researcher-owned and falsifiable by data.*

Assumption A.3 (Sampling)**.** *The unit-by-measure score matrix is drawn under a stated sampling model (e.g., exchangeable country-level aggregates), frozen with the run.*

Assumption A.4 (Tolerance policy)**.** *$\delta \geq 0$ is declared ex ante (default 0) and is never tuned to rescue an empty set.*

A.3 Consistency of the plug-in objects

Proposition A.1 (Consistency of the admissible set and range)**.** *Assume A1–A4. Suppose*

$$\max_{m \in \mathscr{M}} \max_{r \in \mathscr{R}} \big| \hat{s}_r(m) - s_r(m) \big| \xrightarrow{p} 0, \qquad \max_{m \in \mathscr{M}} \big| \hat{\beta}(m) - \beta(m) \big| \xrightarrow{p} 0,$$

and suppose separation: the slack margin $\mu(m) := \min_r s_r(m) + \delta$ satisfies $\mu(m) \neq 0$ for all $m \in \mathscr{M}$. Then $\Pr(\hat{\mathscr{M}}^* = \mathscr{M}^*) \to 1$ and $[\hat{L}, \hat{U}] \xrightarrow{p} [L, U]$.

Proof sketch. Let $\eta := \min_m |\mu(m)| > 0$ by separation. Uniform consistency gives $\Pr(\max_{m,r} |\hat{s}_r - s_r| < \eta/2) \to 1$. On that event, $\hat{\mu}(m)$ has the sign of $\mu(m)$ for every m, so $\hat{\mathscr{M}}^* = \mathscr{M}^*$; on the same event the endpoints are extrema of $\hat{\beta}$ over the fixed finite set $\mathscr{M}^*$, and the asserted convergence follows from the uniform bound on $\hat{\beta}$. Boundary measures with $\mu(m) = 0$ are the only obstruction: their admission is decided by sampling noise. □

Remark A.1 (Empty sets)**.** *$\mathscr{M}^* = \emptyset$ rejects the pair $(\mathscr{M}, \mathscr{R})$, not the construct C; it is a finding, not an error.*

Remark A.2 (Consistency is not validity)**.** *Proposition A.1 says the plug-in objects are well-posed. Validity claims come from the restrictions themselves (A2) and from the informativeness of $\mathscr{B}^*$.*

A.4 Holdout null and claim

Assumption A.5 (Pre-data discipline)**.** *The partition $(\mathscr{R}_S, \mathscr{R}_H)$, tolerance δ, baseline rule B, and random seed are fixed before any holdout data are examined.*

Proposition A.2 (Null and claim)**.** *Under pre-data discipline, $\mathscr{M}_S^*$ does not depend on $\mathscr{R}_H$. Let $H(m) = \mathbf{1}\{m \text{ passes every } r \in \mathscr{R}_H\}$. The claim is $\mathbb{E}[H(m) \mid m \in \mathscr{M}_S^*] > \mathbb{E}[H(m) \mid m \in \mathscr{M}_B^*]$; the null is equality (selection restrictions carry no predictive content for holdout restrictions).*

The natural estimator is the share of survivors passing holdout, $\hat{p} = |\mathscr{M}_S^*|^{-1} \sum_{m \in \mathscr{M}_S^*} H(m)$, compared with the baseline analogue on the same holdout set.

A.5 Coupling and the uncertainty band

The plug-in range inherits sampling error from both $\hat{\beta}(m)$ and $\hat{\mathscr{M}}^*$. When a measure is near the boundary and β-extreme, the two errors are coupled. Endpoint inference for fixed identified sets [29, 47] takes selection as given; here selection is estimated. The main-text protocol—uncertainty band, empty-replicate rate, boundary attribution, optional conservative projection—is an uncertainty summary under a stated reading, not a simultaneous confidence region under arbitrary dependence. Sample splitting decorrelates admission and endpoints at the cost of effective sample size. Formal simultaneous-coverage theory is the subject of the companion methods paper.

A.6 Notation at a glance

Symbol	Meaning
C	Scientific construct
m	Candidate measure / measurement function
$\mathscr{M}$	Declared finite menu
V	Auxiliaries used in validity restrictions
$\mathscr{R}(C)$	Validity restrictions / nomological network
τ_r	Substantive threshold (researcher-owned)
$\mathscr{R}_S, \mathscr{R}_H$	Selection and holdout partitions
$s_r(m)$	Sample slack of restriction r on measure m
δ	Slack tolerance (default 0)
$\mathscr{M}^*$	Admissible set
$\mathscr{M}^*_{\text{select}}, \mathscr{M}^*_{\text{robust}}$	Select and robust sets under holdout designs
$\beta(m)$	Scientific result using measure m
$\mathscr{B}^*$	Admissible-result set (construct-identified)
$[L,U]$	Assumption-indexed admissible-measure range
$G(D,C)$	Measurement-generation technology

B. Synthetic data-generating processes

Synthetic examples remain useful because population truth is known. Consider standardized latent C, a convergent criterion V_c with $\text{Corr}(V_c,C) = 0.80$, a discriminant foil V_d orthogonal to C, an outcome Y with $\text{Corr}(Y,C) = 0.60$, and a known-groups latent gap of 0.80. The menu contains a high-quality measure loading 0.90 on C, a weak measure loading 0.50, a contaminated measure loading 0.70 on C and 0.55 on V_d, a method-variance-dominated measure loading 0.20, and pure noise (Table 4).

Table 4: Population menu for hand-verifiable synthetic examples.

Measure	Loading on C	Loading on V_d	Method variance	Intended role
m_{true}	0.90	0	residual	High-quality operationalization
m_{weak}	0.50	0	residual	Valid but noisy
m_{contam}	0.70	0.55	residual	Contaminated by discriminant foil
m_{method}	0.20	0	dominant	Method-variance dominated
m_{noise}	0	0	1	Negative control

DGP A: clean separation. Selection restrictions are $\text{Corr}(m,V_c) \geq 0.39$, $|\text{Corr}(m,V_d)| \leq 0.25$, and a known-groups gap of at least 0.25; target $\beta(m) = \text{Corr}(m,Y)$. Population slacks are $+0.33$, $+0.01$, $+0.17$, -0.23, -0.39 on convergence; $+0.25$, $+0.25$, -0.30, $+0.25$, $+0.25$ on discrimination; and $+0.47$, $+0.15$, $+0.31$, -0.09, -0.25 on the group gap. Only m_{true} and m_{weak} survive, and

$$\mathscr{B}^* = \{0.54,\, 0.30\}, \qquad [L,U] = [0.30,\, 0.54].$$

Every admissible slack is strictly positive: separation holds.

DGP B: boundary measures and coupling. Raise the convergent threshold to 0.70. Then m_{true} remains admissible by a $+0.02$ margin and m_{weak} is rejected (-0.30); the plug-in range collapses to the singleton $\{0.54\}$. If instead the threshold is set at the weak measure's population correlation (0.40), m_{weak} lies exactly on the boundary: in finite samples its admission is decided by sampling noise, and because it attains the lower endpoint whenever admitted, admission error and endpoint error are coupled—the regime the boundary-attribution protocol monitors.

DGP C: empty $\mathscr{M}^*$ is a finding. Replace the discriminant restriction with $\mathrm{Corr}(m,V_d) \geq 0.50$ and raise the convergent threshold to 0.60. No measure can load heavily on both C (hence on V_c) and the orthogonal foil at once: m_{true} passes convergence ($+0.12$) but fails the foil (-0.50); m_{contam} passes the foil ($+0.05$) but narrowly fails convergence (-0.04). The admissible set is empty—a rejection of the pair $(\mathscr{M},\mathscr{R})$, not of C. The honest response is to revise the menu or the network, never the tolerance δ.

DGP D: holdout discipline. Use a lenient selection screen, $\mathrm{Corr}(m,V_c) \geq 0.40$ and $|\mathrm{Corr}(m,V_d)| \leq 0.60$, and a stricter holdout screen, $|\mathrm{Corr}(m,V_d)| \leq 0.20$ and a group gap of at least 0.25. Selection admits m_{true}, m_{weak}, and m_{contam}; on holdout, m_{contam} fails the tight discriminant test (-0.35) while the other two pass, giving a construct-guided pass rate of $2/3$. An anchor-fit baseline that keeps the two measures with the largest convergent correlations retains m_{true} and m_{contam}, of which only one survives holdout: pass rate $1/2$. In this population-limit example, construct-guided selection beats single-anchor selection on held-out validity—the direction of the falsifiable claim.

A finite-sample engine check under the shipped implementation reproduces the qualitative structure of DGPs A–D; run identifiers are recorded in the companion repository.

C. Benchmarks as measurement menus

A benchmark maps model behavior into a score interpreted as a capability. If several task families purport to measure the same capability, they form a measurement menu, and construct-identified benchmark inference would define the capability and intended use, declare the benchmark families, specify convergent and discriminant evidence, reserve held-out task families, and report model comparisons over the admissible benchmark set [49, 3]. This perspective does not solve contamination or benchmark governance [48], but it clarifies the scientific object: a leaderboard score is an operationalization, not the capability itself. The motivation is empirical: the AI Index reports capability-benchmark reporting by leading developers as near-universal while responsible-AI benchmark reporting remains uneven [21].

D. Validation gates in automated social science

Language models can help generate hypotheses, propose coding schemes, construct candidate measures, and simulate respondents [34]. The design question is which commitments must remain inspectable when the workflow becomes agentic. Automation may propose hypotheses, generate candidates, compute frozen scores, and execute declared restrictions; it must not silently choose the construct definition, rewrite thresholds after observing outcomes, or validate one model with another model's correlated outputs as though they were independent evidence. Interventions in LLM-simulated-user experiments can induce user drift, motivating negative-control outcomes as a diagnostic [28]; prompt-engineering and fine-tuning repairs must be distinguished from statistical calibration using auxiliary human data under explicit assumptions [27]; and early case studies find that frontier agents complete engineering work without substantially answering the research questions [32]. The most direct Goodharting failure is training a generator against the known restriction network; the defenses are procedural—freeze $\mathscr{R}$ and τ before final scoring, preserve negative controls, reserve restrictions or units, and keep generator provenance separate from validity evidence. The medium-run opportunity is a community registry of construct definitions, frozen score matrices, researcher-authored networks, and reproducible audits—a way to make disagreements legible rather than a closed "verifiable reward" for autonomous science.

E. Open-weight LLM measurement protocol and appendix pilot

LLM outputs enter the framework only after they are reduced to declared score columns. For a binary or ordered response prompt, a log-probability scalarization uses the model-implied probability or log-odds assigned to prespecified answer tokens. The prompt template, model identity, quantization, tokenizer, decoding settings, answer-token mapping, and scalar reduction are part of the measurement function m and must be frozen with the score provenance. Temperature-zero decoding improves reproducibility but does not establish validity; open weights improve auditability but do not confer scientific standing.

The 2026-08-10 preliminary patience freeze (shipped under `cvprofiles` 3.0.0; seed 20260810, split seed 17, $n = 41$) used two open-weight models: Meta-Llama-3.1-8B-Instruct Q8_0 ($m_{\text{prompt\_a}}$) and Phi-4-mini 3.8B Q8_0 ($m_{\text{prompt\_b}}$), GGUF-pinned by checksum, scored with llama.cpp at temperature 0. Table 5 states the required provenance record.

The 2026-08-10 patience pilot. In the pooled $K = 5$ country units-split pilot, the selection set across all folds was $\mathscr{M}^*_{\text{select}} = \{m_{\text{gps\_patience}}, m_{\text{prompt\_a}}\}$, with range $[L,U] = [0.328, 0.402]$ on the standardized log GDP per capita estimand with education control. However, the robust set complying on all five held-out test frames was empty ($\mathscr{M}^*_{\text{robust}} = \varnothing$).

Table 5: Minimum adapter card for an LLM measurement function.

Field	**Required record**
Construct prompt	Exact text and response format.
Model	Repository/model identifier and pinned artifact hash.
Runtime	Inference engine, quantization, and tokenizer.
Decoding	Temperature, seed, and any sampling parameters.
Scalarization	Exact map from token probabilities/outputs to the score column.
Provenance	Whether prompts/models were screened before the menu freeze; all exclusions.
Independence	Any overlap between training data, validity auxiliaries, and downstream outcomes.

Moreover, the fold-0 monotonicity slack of GPS patience was -0.383 (against threshold 0.15), which our full-sample analysis in Section 5.4 demonstrated was a small-sample training-split artifact (the full 41-country monotonicity slack is $+0.233$). Under units-resampling bootstrap, the fold-0 uncertainty band $[0.101, 0.561]$ swallows the plug-in range. This confirms why the $K = 5$ mixed-menu freeze is retained as an illustrative pilot of the protocol rather than a confirmatory scientific conclusion.

Table 6: Appendix pilot: 2026-08-10 frozen patience run (41 countries, pooled K=5, `cvprofiles` 3.0.0). Slacks are on the fold-0 training frame; β is pooled standardized OLS on log GDP per capita with education control; selection and compliance are counts out of 5 folds.

Measure	s_{conv} (θ=0.20)	s_{mono} (θ=0.15)	s_{disc} (θ=0.35)	β (pooled)	**Selected** (of 5)	**Compliant** (of 5)
$m_{\text{gps\_patience}}$	$+0.298$	-0.383	$+0.028$	$+0.4025$	5	1
$m_{\text{wvs\_q13}}$	-0.453	-0.317	$+0.143$	-0.1660	0	1
$m_{\text{wvs\_q14}}$	-0.765	-0.717	$+0.300$	-0.1948	0	0
$m_{\text{composite}}$	-0.678	-0.650	$+0.266$	-0.2187	0	0
$m_{\text{prompt\_a}}$	$+0.153$	$+0.433$	$+0.326$	$+0.3275$	5	3
$m_{\text{prompt\_b}}$	-0.059	-0.017	$+0.234$	$+0.3739$	1	1
m_{noise}	-0.342	$+0.100$	$+0.231$	$+0.0199$	0	1

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